Project Name: LunchSpot

Team Members:

1. Putra Jadi Mukti
2. Fasya Raihan Maulana
3. Zulfa Fatah Akbar Ahmad

Problem (The "Why")

- **Current Issue:** People walk to the canteen just to check if there's a free spot. Half the time it's packed, and the trip was for nothing.
- **Impact:** Wasted trips cost time and energy, especially during peak hours when breaks are short and every minute matters.

Proposed Solution

- **Our Approach:** Use computer vision to monitor canteen occupancy in real time and show people whether a spot is full or open before they leave their desk or room.
- **Key Value:** People check their phone instead of walking over to look. No more wasted trips to a full canteen.

Core Features

- **Feature 1:** Real-time occupancy status per canteen, detected through computer vision and shown as full or available.
- **Feature 2:** Occupancy history, so people can spot patterns like which hours tend to be busiest.
- **Feature 3:** Alert notification when a full canteen frees up, instead of a fixed daily reminder. This fits the use case better since people care about the moment a spot opens, not a scheduled check-in.

High-Level Architecture & Quality Assurance

- **Architecture Pattern:** Clean Architecture strictly adhering to SOLID principles.
- **Design Patterns Used:** Repository Pattern, Dependency Injection
- **Code Quality Goal:** Must Have 0 warnings as possible (scanned via static analysis), and 100 % Unit Test Coverage.

Identified Challenges & Risks

- **Challenge:** Data lag. The camera detects a seat as open, but by the time someone walks over, another person has taken it. The gap between detection and arrival can make the app look wrong.
- **Challenge:** ML identifies tables that appear in various forms, such as circular tables or tables that merge into one.
- **Risk:** Computer vision accuracy. Poor lighting, camera angle, or crowded frames can throw off seat detection and hurt trust in the app.